ENERGY STORAGE FIRE & SAFETY FACT SHEET

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As battery energy storage systems (BESS) and other clean technologies become more common, public concern over fire risk has grown, often fueled by misunderstanding or outdated data. Let's take a clear, evidence-based look at fire safety in energy storage systems, comparing real-world incidents, highlighting improvements, and addressing the most common misconceptions.

BESS and other renewable systems are **significantly more safe** then our fossil fuels energy systems of pumps, pipelines, refineries, plants, and residential/commercial structures. Furthermore, the widespread adoption and installation of BESS directly reduces the need to rely on these systems and provides many additional practical benefits.

Whereas clean energy systems simply require the source of generation (wind, solar, hydropower, etc) a place to store that energy (BESS), and distribution infrastructure, fossil fuel systems require a more complex and messy logistics pipeline of extraction, refinement, and transport that leaves multiple vulnerabilities to fire and other accidents.

Every year, on average, the US experiences:

286
Oil Pipeline Incidents[1]

150
Oil Spills[2]

11
Oil Refinery Incidents[3]

125,000
Gas Leaks[4]

27,900
Home Heating Fires[5]

167,800
Home Cooking Fires[5]

All together, these incidents directly result in:

254 Deaths

3,084 Injuries

$2.5 Billion+ in Damage

ADDITIONAL RISKS:
A rigorous study found living next to an oil refinery results in a **10% Increased risk of cancer**[6]

In NYC alone, **22,000 gas leaks** occurred, often caused by poor or decaying infrastructure within buildings.[7]

Harvard estimates **1 in 5 deaths** occur due to air pollution from fossil fuels worldwide.[8]

While not prone to overt disaster, the particulate matter released from coal power plants has been responsible for over **460k deaths** since 2003.[9]

In addition to this data, it is important to be aware that many fossil fuel incidents are often **large, explosive, and cause immediate tragic loss of life and property** even in the best cases. BESS and clean energy incidents, in addition to being dramatically more rare, do not compare in devastation to property and people.

How do BESS systems compare?

BESS technology is rapidly evolving in its safety and stability. Single incidents often immediately trigger legislation, regulations, and adjustments that are quickly implemented in future energy storage applications. This rate of progress is so significant in fact, battery storage failure incident rate dropped 97% between 2018 and 2023[10]. This demonstrable rate of progress in safety is simply not present in fossil fuel energy, because of the inherent physics challenges of flammable oil & gas.

2018 | 9.2 faults per GW

2023 | 0.2 faults per GW

- 65% of these faults were due to improper operation and integration, only 11% were due to actual failures of the battery cell/module.[10] This further emphasises the need for proper workforce training in BESS.
- New national energy storage safety standards such as **UL 9540**[11], **UL 9540A**[12] and **NFPA 855**[13], add rigorous protocols to new BESS systems.
- A **Model Energy Storage Ordinance Framework** was released in June 2025 by the American Clean Power Association that address the issues present in older energy storage systems.[14]
- New real-time Battery Management Systems (BMS) catch thermal runaway events **50-80 cycles** before they occur.[15]
- NYSERDA'S Quality Assurance team has conducted **57 in-field inspections** of BESS facilities >300 kW, leading to enhanced facility oversight, inspection process, and code recommendations.[15]

Let's observe some example incidents.

ANATOMY OF AN INCIDENT

Let's take a look at two New York incidents, one due to fossil fuels and the other from energy storage:

2014 East Harlem Gas Explosion[16]

- Caused by leaking natural gas pipeline under a residential street
- Explosion destroyed two apartment buildings instantly
- **8 fatalities, 50+ injuries**, many displaced
- Fires raged for hours, widespread debris and property damage
- Actual steps to future prevention of further incidents is murky and currently unknown

2023 East Hampton BESS Failure[17]

- Triggered by overheating module in a utility-scale lithium-ion battery system
- Automatic safety systems detected issue and initiated safe shutdown
- No deaths or injuries, minimal damage
- Contained to the storage unit; no explosive fuel or fire spread
- Prompted direct and effective legislation and regulation that prevents this event from happening in the future

This comparison underscores a critical point: while fossil fuel accidents often produce catastrophic explosions, fatalities, and widespread damage, modern BESS systems are designed with containment and fail-safes that minimize risk.

What systems in BESS stop fires?

Fire Prevention Systems[18]

Battery Management Systems (BMS)
Keep cells within safe temperature ranges.

Metal Separators & Flame-Retardant Cell Fill
Slow heat and fire spread.

Voltage and Temperature Sensors
Detect abnormal conditions early.

Automatic Disconnects/Fuses
Isolate failing cells or racks.

Steel & Fire-Rated Enclosures
Surround the entire system.

Fire gaps & Firewalls
Contain runaway events.

Pressure Relief Vents
Safely release any build-up of gases.

Smoke, Heat, and Gas Detectors
Identify incipient thermal runaway.

Automated System Shutdown Protocols
Activate before fire develops.

Remote Monitoring & Control
Allows human operators to depower/isolate system

Pre-Planned Emergency Procedures
Coordinated with local fire departments.

Fire Suppression Systems[18]

Clean-Agent Gas
Releases inert gas to extinguish fires.

Water Mist & Sprinkler Deluge
Releases water & mist.

Aerosol or Dry-Chemical
Material found in handheld fire extinguishers

Battery energy storage systems (BESS) and other clean energy technologies operate under fundamentally different safety assumptions than fossil fuel systems. Despite being new technology, Modern BESS units are already rapidly growing in safety year by year, and regulations with protocols are now in place to stop incidents. Early detection, containment, and automated shutdowns to isolate incidents before they can escalate are now the standard.

Ultimately BESS is a safe, reliable, and lasting energy storage option.

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